Guard cell size and pore aperture influence stomatal closure kinetics

Christopher D. Muir[1,2*] Wei Shen Lim[1]

[1]School of Life Sciences, University of Hawai‘i; Mānoa, Hawai‘i, USA
[2]Department of Botany, University of Wisconsin, Madison; Wisconsin, USA

*Corresponding author: cdmuir@wisc.edu

ORCID: Christopher D. Muir — 0000-0003-2555-3878

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notes

Kirschbaum et al. 1988 - read

Tulva et al. (2025) - not super relevant for this paper.

Hsu et al. (2021) - t50 WT aribodopsis: 5.1 ± 0.4 min. Their VPD method “followed by □2.0 kPa VPD (20 to 32% relative air humidity) in most experiments, as indicated in the figures”. There was a lot in this paper. Basically, they don’t know exactly how VPD is sensed, but show that it’s independent of CO2 mechanism.

Some older references:

(this measured steady state responses, not kinetics) Morison J.I.L. Gifford R.M. 1983. Stomatal sensitivity to carbon dioxide and humidity. A comparison of two C3 and C4 grass species. Plant Physiol. 71:789–796.

(not really relevant) Talbott L.D. Rahveh E. Zeiger E. 2003. Relative humidity is a key factor in the acclimation of the stomatal response to CO2. J. Exp. Bot. 54:2141–2147.

hall_leaf_1976 - shows that gsw responds to VPD per se at constant LWP

Bunce (1997) - not very relevant

Blatt et al. (2026) - not sure if this is useful. Shows that guard cell walls stay hydrated even when airspace is subsaturated.

Buckley 2016 (10.1111/pce.12651) provides useful background on why hydroactive response takes time (ABA synthesis).

Merilo et al. (2018) say "The ABA signaling unit consists of cytosolic PYRABACTIN RESISTANCE1 (PYR1)/PYR1-LIKE (PYL)/ REGULATORY COMPONENTS OF ABA RECEPTORS (RCAR) receptors that, in the presence of ABA sequester type 2 protein phosphatases, lead to the activation of protein kinase OST1, which, in turn, activates the slow-type anion channel SLAC1 and triggers stomatal closure (Geiger et al., 2009; Lee et al., 2009; Ma et al., 2009; Park et al., 2009)" (these are steps after ABA synthesis)

Ivanoff 1928 - original wrong-way response citation? from Jalakas et al 2021 ("promoting a transient passive 'wrong-way' stomatal opening (Ivanoff, 1928; Mott & Franks, 2001; Buckley, 2019).")

FYI, t50 = "6.1 ± 0.32 min in Arabidopsis Col-0 wild-type (Merilo et al., 2018)" I also have read this: https://nph-onlinelibrary-wiley-com.ezproxy.library.wisc.edu/doi/full/10.1111/nph.17592

Need to check out these references "Although the plant hormone abscisic acid (ABA) is involved in stomatal closure in response to drought stress and reduced air humidity (Bunce, 1997; Roelfsema & Hedrich, 2005; Bauer et al., 2013), the existence of an ABA-independent pathway participating in low humidity-induced stomatal response has been suggested (Xie et al., 2006). Furthermore, stomata may also directly respond to leaf water status without relying on ABA production and signalling (see Buckley, 2005; Damour et al., 2010; Peak & Mott, 2011). However, both the metabolically mediated hydro-active feedback response of guard cell osmotic pressure to the water status of epidermal evaporative site (Buckley, 2005) and the stomatal response mechanism based on a near-equilibrium in water potential between the guard cells and the air in the stomatal pore (Peak & Mott, 2011) still embed the potential for ABA to be involved in air humidity-related stomatal behaviour."

Binstock et al. (2024) 10.1093/plphys/kiad560 - see if they have estimates of tau

This paper looked at epidermal cell : guard cell size in domesticated tomato and related to closure kinetics. See if I can do this? 10.1111/ppl.70543 (use Sack and Buckley estimate of epidermal cell size based on stomatal index, density, and guard cell size)

NOT THAT USEFUL - Tinoco-Ojanguren C, Pearcy RW. (1993) Stomatal dynamics and its importance to carbon gain in 2 rain forest Piper species. 1. VPD effects on the transient stomatal response to light flecks. Oecologia 94: 388–394

Some OG citations "Changes in stomatal aperture are driven by changes in turgor pressure of guard cells (Heath 1938) due to the accumulation of ions and/or solutes (Imamura 1943; Fujino 1967; Outlaw 1983), which increases the osmotic potential and lowers the water potential of guard cells (Weyers and Meidner 1990; Willmer and Fricker 1996)." quote from Lawson et al (2010)

Abstract

Guard cell size and stomatal density vary over X and X orders of magnitude among extant vascular land plants, yet the adaptive significance of much of this variation remains unclear. The evolution of guard cell size in particular is poorly understood and may even be constrained by nonadaptive features like genome size. One hypothesis is that natural selection may favor smaller guard cells to increase the rate at which stomatal conductance responds to fluctuating environmental conditions. A related hypothesis is that operational stomatal conductance (g_{op}) is often $\approx 25\%$ of its theoretical maximum (gmax) because at this stomatal aperture, guard cell volume is poised to change rapidly with changes in turgor pressure. The support for both hypothesis is limited and mixed, in part because they have not been tested together, even though both guard cell size and the ratio of g_{op} to g_{max} (f_{gmax}) should influence stomatal kinetics. We analyzed 2,124 estimates of stomatal closure kinetics in response to an abrupt increase in vapor pressure deficit (VPD) among 29 diverse wild tomato populations in the genus *Solanum*. Both smaller guard cell size and lower f_{gmax} were associated with faster stomatal closure kinetics, but at different levels of biological organization. Guard cell size explained X% of the variance in stomatal kinetics among populations, whereas gop:gmax explained X% of the variance among individuals within populations. [neither result would have been supported if analyzed in isolation - is this true?]. We conclude that guard cell size may respond to selection on stomatal kinetics over evolutionary time, but actual kinetics are modulated by a tradeoff between maximizing gop and minimizing the time required for stomatal closure. Putting these hypotheses together with stabilizing selection on gop may explain why stomatal size and density negatively covary among species.

Introduction

In nature, change is the only constant. One way that vascular plants (tracheophytes) cope with change is by adjusting stomatal pore aperture to optimize the trade-off between carbon gain and water loss (1–3). In the absence of physical limits and energetic tradeoffs, natural selection would favor plants that could instantaneously adjust stomatal conductance (g_{sw}) to perfectly track dynamic environmental conditions (4). In reality, stomatal responses take time, creating a lag between actual and optimal stomatal aperture. For example, many species use hydroactive feedback mechanisms to precisely control stomatal responses to changing vapor pressure deficit (VPD), the driving force for evaporation (5). Hydroactive stomatal control requires sensing a change in VPD through as-yet-unknown mechanisms, signal transduction, and efflux of osmotic solutes to alter guard cell turgor more than the surrounding epidermal cells. Abscisic acid (ABA) dependent responses require several minutes to accumulate or synthesize hormone pools *de novo* (6–9); ABA-independent stomatal responses require changes in gene expression and protein synthesis that take time (10). Once anion channels are activated, the rate of change in stomatal aperture depends on the rate of change in guard cell turgor (dP_{g}/dt) and the sensitivity of aperture to change in turgor ($da_{\text{S}}/dP_{\text{g}}$). As we discuss below, these two factors (dP_{g}/dt and $da_{\text{S}}/dP_{\text{g}}$) may be influenced by guard cell size and the ratio of operational to maximum stomatal conductance (f_{gmax}), respectively.

Over a century ago (11) observed that after an initial increase, leaf transpiration decreases in response to decreasing humidity. He correctly attributed changes in transpiration to changes in stomatal aperture and subsequent studies have confirmed that stomatal aperture responds to humidity (12) and, more specifically, transpiration rate (13, 14), which is the product of VPD and g_{sw}. The VPD is related to RH but also temperature. In angiosperms and some pteridophytes (15–17), a sudden increase in VPD, usually induced by lower RH at a constant temperature, causes a transient increase in g_{sw}, as Darwin observed. After a period of several minutes, stomata close until the leaf reaches a new steady-state g_{sw}. These phases of stomatal response to VPD are often referred to as the "wrong-way response" (WWR) and "right-way response" (RWR), respectively. The normative 'wrong' and 'right' appellations refer to the presumed adaptive significance. At steady state, g_{sw} is negatively correlated with VPD, all else being equal, which is interpreted as an adaptive response to the marginal carbon cost of water (1, 18, 19). Models built around this assumption generally predict steady state g_{sw} well, although the functional form and presumed fitness costs are areas of active research (20–24).

Stomatal responses are as much about the journey (kinetics) as they are about the destination (steady state). If stomatal response kinetics are too slow relative to the temporal grain size of environmental variability, then it would be futile for g_{sw} to track environmental variation since the leaf will constantly lag the environment, perpetually fighting the last war. This may explain why g_{sw} can be insensitive to short sun/shade flecks (25) and shaded leaves are likely primed to cope with the hydraulic demands of sudden increases in light, temperature, and VPD (26). If stomatal responses can track environmental variation, then minimizing the lag time can greatly improve resource-use efficiency (4). For example, optogenetic control of stomatal aperture in response to fluctuating light increases water-use efficiency in *Arabidopsis* (27). Therefore the rate of stomatal opening and closing is predicted to have a major effect on how natural selection optimizes stomatal control in variable environments (28).

Much of the research on stomatal kinetics focuses on light responses because light intensity is usually the most important, rapidly fluctuating environmental variable affecting photosynthesis. In nature, fluctuating light intensity simulateneously alters leaf temperature and VPD (e.g. 29). We focus on stomatal closure kinetic in response to an abrupt increased in for three reasons. First of all, the primary aims and *a priori* hypotheses of this study were not to study stomatal kinetic parameters and are described in (30). The idea to opportunistically use g_{sw} response curves generated during this study to test hypotheses about which traits influence stomatal kinetics came after the data had been collected, but not yet analyzed for this purpose. Second, VPD responses exhibit a similar functional form as light-responses, suggesting similar underlying mechanisms (5). Factors that affect stomatal response kinetics to VPD likely generalize to other environmental variables. Third, humidity responses tend to dominate over and are faster than other environmental responses (31, 32), but there is variation among species (33). Thus, factors that constrain maximum rates of stomatal closure are most likely to be observed in response to VPD. Finally, the ecological salience of VPD responses is increasing as climate change warms the atmosphere (34).

Fully describing stomatal responses to the environment is complex (9, 35, 36), but the dynamics of g_{sw} in response to an abrupt environmental change is well described by relatively simple phenomenological functions (37, 38). A four-parameter equation describes stomatal response kinetics to a step change in light intensity across dozens of plant species (39):

$$g_{\mathrm{sw,t}} = g_{\mathrm{f}} + (g_{\mathrm{i}} - g_{\mathrm{f}})e^{-\left(\frac{t}{\tau}\right)^{\lambda}}. \tag{1}$$

In this equation, $g_{\mathrm{sw,t}}$ is stomatal conductance at time t, g_{i} is the initial g_{sw} before the step change, g_{f} is the final g_{sw} after the step change, τ is a time constant that describes how quickly stomata respond to the step change, and λ is a shape parameter that describes how the response rate changes over time. When $\lambda = 1$, the response rate is constant over time and τ represents the time required for g_{sw} to reach 36.8% of the way between initial and final values. When $\lambda > 1$, the response rate increases over time, and when $\lambda < 1$, the response rate decreases over time. (39) refer to λ as a lag-time parameter because empirically $\lambda > 1$. We refer to stomatal kinetic parameters τ (time-constant) and λ (lag-time) henceforth. The first goal of this study is to evaluate whether kinetics of the RWR after a step change in VPD are adequately described by Equation 1. If so, it suggests that some fundamentally similar mechanisms are shared between light and hydroactive stomatal responses (40).

The main goal of this study is to test hypotheses about which stomatal anatomical traits affect kinetic parameters τ and λ. We consider two interconnected hypotheses that have thus far been treated in isolation. The first hypothesis is that smaller guard cells open and close faster because of their intrinsically greater surface area to volume ratio (39, 41–43). For approximately cylindrical cell geometries, like that found in guard cells, surface area increases linearly with radius, whereas volume increases in proportion to the radius squared. Consequently, larger guard cells will require more time, all else being equal. to achieve a given change in turgor pressure and, hence, stomatal pore aperture. A plant with large guard cells can partially compensate for this geometric constraint by increasing the activity and/or density of ion and solute transporters in the guard cell plasma membrane (44–46). However, this may be limited by the available membrane area and/or energetic costs of maintaining high transporter densities (37). Empirical support for the hypothesis that smaller guard cells are faster is mixed and has primarily come from step changes in light (38, 42, 47–51). Meta-analysis of dozens of species suggests that stomatal size alone is weakly related to kinetics and modulated by whether guard cells are kidney or dumbbell shaped (39). The relationship between guard cell size and stomatal response kinetics to leaf water potential (52) and VPD (53, 54) is less well studied, but similarly equivocal.

One factor that might complicate the relationship between size and speed is aperture. Stomatal aperture can vary from near 0 when guard cell turgor pressure is low and asymptotically approach g_{max} as guard cell turgor pressure approaches ∞ (Figure 1X). The relationship between guard cell turgor pressure and stomatal aperture is nonlinear (55) as cell walls become less elastic at greater turgor (56). The nonlinear relationship between guard cell turgor and pore aperture implies that if the rate of turgor change is constant ($\frac{\mathrm{d}P}{\mathrm{d}t} = C$), the rate of change in aperture, and hence stomatal conductance, will vary depending on the initial aperture (a_0) relative to the maximum aperture (a_{max}). Hence, $\frac{\mathrm{d}a}{\mathrm{d}t} = f(a_0/a_{\mathrm{max}})$. When initial aperture is high, g_{sw} will respond more slowly than when initial aperture is low. Since we generally do not observe individual stomatal aperture, relating it this idea to leaf-level g_{sw} requires scaling by stomatal density, biophysical constants, and morphological constants (e.g. 57). We will use the term "fraction of anatomical maximum stomatal conductance", symbolized as f_{gmax} (this is denoted as g_{ratio} by (58)). This value should be proportional to the average aperture divided by its maximum aperture for any arbitrary stomatal density. Leaves operating at f_{gmax} closer to unity

will, all else being equal, will be slower to respond than leaves operating with a f_{gmax} close to zero, independent of stomatal density. (59) hypothesized that selection on stomatal density would maintain typical operational stomatal conductance (g_{op}) relative to g_{max} at a sufficiently low value that g_{sw} would be sensitive to relatively small changes in guard cell turgor pressure. Consistent will this hypothesis, the g_{op}:g_{max} ratio is often near 0.25 (60, 61), well below $f_{\text{gmax}} = 1$, and in a range where g_{sw} would be responsive to small changes in guard cell turgor pressure. There is likely variation within and among species and growth forms in the g_{op}:g_{max} ratio (58, 62).

Putting these two hypotheses together, we predict that both guard cell size and f_{gmax} will influence stomatal kinetics, but possibly at different scales of biological organization. Guard cell size tends to vary less than stomatal density or aperture at many biological scales. For example, on a mature leaf, guard cell length is nearly constant even as cell volume and pore aperture change with turgor (56, 63). Compared to stomatal density, guard cell size can be less variable (64, 65) and evolve more slowly between species (66). If there is relatively little plastic or genetic variation among individuals within a species, then most of the variance in stomatal kinetics within species cannot be explained by variation in stomatal size. In contrast, f_{gmax} can vary immensely within and among individuals because stomatal aperture responds dynamically over the day and in response to environmental variation. Because guard cell size and aperture typically vary at different levels of biological organization, we predict that guard cell size likely explains more variation in stomatal kinetics among species, whereas f_{gmax} will explain more variation within species. Within species variation can arise from either genetic differences among individuals or environmental plasticity. We will also test faster responses of adaxial stomata respond (39, 67, 68) can be explained differences in guard cell size and/or f_{gmax} between surfaces.

One challenge in determining whether traits covary at higher levels of organization is that statistical procedures to account for phylogenetic nonindependence can obscure ecologically relevant covariation caused by natural selection (69, 70). For example, in a meta-analysis of stomatal kinetic responses to light, the explanatory power of traits like guard cell size and stomatal ratio declined in phylogenetic compared to nonphylogenetic regression (39). This finding may be caused in large part because grasses have small stomata, high adaxial:abaxial stomatal ratio, and rapid stomatal kinetics. Are these associations a coincidence caused by shared ancestry or ecologically important covariation that helps explain how grasses adapt to their environment? Recent advances in phylogenetic comparative methods use hierarchical, multiresponse phylogenetic models (66, 71, 72) to tease apart "conservative trait correlations" that act at higher levels of organization (e.g. among species) from trait correlations that explain variation at lower levels of organization (e.g. among individuals within species and plastic responses). We use this approach to test whether guard cell size and f_{gmax} explain variation in stomatal kinetics at different levels of biological organization.

The four main questions we address in this study are:

1. Are VPD kinetics adequately characterized by the same functional form as light responses?
2. Are l_{gc} and f_{gmax} positively associated with τ?
3. At what biological levels of organization do l_{gc} and f_{gmax} vary and influence τ?
4. Are abaxial-only stomatal responses slower than whole leaf stomatal responses?

As noted above, we are opportunistically analyzing data from an experiment designed to answer different questions. Before analyzing the kinetic data, we planned to test whether 1) guard cell size would influence τ and 2) that τ would be greater on the abaxial-only leaves based on a recent metaanalysis of stomatal responses to light (39). We made the decision to test for an association between f_{gmax} and τ after exploratory analyses revealed a potential association. Adding hypotheses after some results have been analyzed, also known as "Hypothesizing After Results Known" (HARKing), can lead to reports of spurious associations that are not repeatable in later studies (73, 74). Therefore, our conclusions about the effect of f_{gmax} on stomatal kinetics should be interpreted with caution until they are replicated.

Materials and Methods

Populations and phylogeny

We analyzed stomatal responses to changing humidity among 29 ecologically diverse populations of wild tomato, including representatives of all described species of *Solanum* sect. *Lycopersicon* and sect. *Lycopersicoides* (75) and the cultivated tomato *S. lycopersicum* var. *lycopersicum* (Table 1), as described in (30). These populations were selected because they encompass the breadth of climatic variation in the wild tomato clade (76). In one case, we substituted a congeneric population of the species (*S. galapagense*) because of difficulty growing the focal population and we also added a population of *S. pennellii*. Due to constraints on growth space and time, we spread out measurements over -Inf weeks from Inf to -Inf. Replicates within population were evenly spread out over this period to prevent confounding of temporal variation in growth conditions with variation among populations. To estimate phylogenetic effects, we used the RAxML whole-transcriptome concatenated phylogeny based on 2,745 100-kb genomic windows from (76). Two of our populations were not in this tree. We used accession LA1044 in place of LA3909, two populations of *S. galapagense*; LA0750 was added as sister to LA0716, two closely related populations of *S. pennellii*. The node separating LA0750 from LA0716 was placed half-way between the next deepest node.

Table 1: Accession information of *Solanum* populations used in this study. The species name, accession number, collection latitude, longitude, and elevation. TGRC: Tomato Genetics Resource Center.

Species	TGRC accession	Latitude	Longitude	Elevation (mas)
S. arcanum	LA2172	-6.008	-78.858	662
S. cheesmaniae	LA0429	-0.644	-90.329	800
S. cheesmaniae	LA3124	-0.804	-90.042	1
S. chilense	LA1782	-15.267	-74.633	1000
S. chilense	LA4117A	-22.907	-67.941	3540
S. chmielewskii	LA1028	-13.883	-73.017	3000
S. chmielewskii	LA1316	-13.400	-73.906	2920
S. corneliomulleri	LA0107	-13.117	-76.383	60

(continued)

Species	TGRC accession	Latitude	Longitude	Elevation (mas)
S. corneliomulleri	LA0444	-13.433	-76.133	100
S. galapagense	LA0436	-0.953	-90.978	40
S. galapagense	LA1044	-0.284	-90.548	0
S. habrochaites	LA0407	-2.181	-79.884	70
S. habrochaites	LA1777	-9.550	-77.700	3216
S. huaylasense	LA1358	-9.533	-77.967	750
S. huaylasense	LA1360	-9.546	-77.929	1490
S. huaylasense	LA1364	-10.133	-77.383	2920
S. lycopersicoides	LA2951	-19.317	-69.450	2200
S. lycopersicoides	LA4126	-19.287	-69.396	3120
S. neorickii	LA1322	-13.483	-72.442	2380
S. neorickii	LA2133	-3.400	-79.183	1980
S. pennellii	LA0716	-16.225	-73.617	50
S. pennellii	LA0750	-14.775	-75.034	550
S. pennellii	LA3778	-14.775	-75.034	616
S. peruvianum	LA2744	-18.550	-70.150	400
S. peruvianum	LA2964	-18.028	-70.835	75
S. pimpinellifolium	LA1269	-11.483	-77.075	400
S. pimpinellifolium	LA1589	-8.433	-78.817	30
S. pimpinellifolium	LA2933	-1.442	-80.562	375
S. sitiens	LA4116	-22.159	-68.782	2960

Plant growth conditions and light treatments

A thorough description of plant growth conditions can be found (30), therefore we summarize the most important information here. Seeds provided by the Tomato Genetics Resource Center germinated on moist paper in plastic boxes after soaking for 30-60 minutes in a 50% (volume per volume) solution of household bleach and water, followed by a thorough rinse. We transferred seedlings to cell-pack flats containing Pro-Mix BX potting mix (Premier Tech, Rivière-du-Loup, Quebec, Canada) once cotyledons fully emerged, typically within 1-2 weeks of sowing. We grew seeds and seedlings for both sun and shade treatments under the same environmental conditions (12:12 h, 24.3:21.7 °C, 49.6:58.4 RH day:night cycle). LED light provided PPFD $= 267\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ (Fluence RAZRx, Austin, Texas, USA) during the germination and seedling stages.

Immediately prior to starting growth light intensity treatments (sun and shade), we transplanted seedlings to 3.78 L plastic pots containing 60% Pro-Mix BX potting mix, 20% coral sand (Pro-Pak, Honolulu, Hawai‘i, USA), and 20% cinders (Niu Nursery, Honolulu, Hawai‘i, USA) withslow release NPK fertilizer following manufacturer instructions (Osmocote Smart-Release Plant Food Flower &

Vegetable, The Scotts Company, Marysville, Ohio, USA). Percentage composition is on a volume basis. We watered to field capacity three times per week to prevent drought stress. Seedlings were randomly assigned in alternating order within population to the sun or shade treatment during transplanting. The average daytime PPFD was $761\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ and $115\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ for sun and shade treatments, respectively.

Humidity response curves

We selected a fully expanded, unshaded leaf at least six leaves above the cotyledons during early vegetative growth. This typically meant that plants had grown in light treatments for ≈ 4 weeks, ensuring they had time to sense and respond developmentally to the light intensity of the treatment rather than the seedling conditions (77). See (30) for further detail.

To measure stomatal responses to a step change in humidity at a constant leaf temperature, which we refer to this as a humidity-response curve. We measured humidity-response curves using a portable infrared gas analyzer (LI-6800PF, LI-COR Biosciences, Lincoln, Nebraska, USA). Light-acclimated plants were placed under LEDs dimmed to match their light treatment during gas exchange measurements. We collected four humidity-response curves per leaf, an amphi (untreated) curve and a pseudohypo (treated) curve at high light-intensity (PPFD $= 2000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$; 97.8:2.24 red:blue) and low light-intensity (PPFD $= 150\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$; 87.0:13.0 red:blue). We always measured high light-intensity curves first because photosynthetic downregulation is faster than upregulation in these species. Pseudohypostomatous leaves are the same as the amphistomatous leaves except gas exchange through the upper (adaxial) surface blocked by a neutral density plastic (propafilm). Comparing amphi and pseudohypo curves on the same leaf enables us to test whether stomatal kinetics are different for ab- and adaxial stomata on the same leaf. To compensate for reduced transmission, we increased incident PPFD for pseudohypo leaves by a factor 1/0.91, the inverse of the measured transmissivity of the propafilm. We also set the stomatal conductance ratio, for purposes of calculating boundary layer conductance, to 0 for pseudohypo leaves following manufacturer directions. To control for order effects, we alternated between starting with amphi or pseudohypo leaf measurements. We made measurements over two days. On the first day, we measured high and low light-intensity curves for either amphi or pseudohypo leaves; on the second day, we measured high and low light-intensity curves on the other leaf type.

In all humidity-response curves, we acclimated the focal leaf ambient CO_2 ($C_a = 415\,\mu\text{mol}\,\text{mol}^{-1}$), $T_{\text{leaf}} = 25.0\,°\text{C}$, high light (PPFD $= 2000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$), and high relative humidity (RH $= 70\%$; VPD $\approx 0.82\,\text{kPa}$) until g_{sw} reached its maximum. After that, we scrubbed water vapor from the incoming air using silica desiccant as quickly as possible, resulting in a water vapor concentration in the incoming air to be on average $0.36\,\text{mmol}\,\text{mol}^{-1}$. This treatment induced rapid stomatal closure, but it did not standardize chamber air humidity because water vapor was introduced by leaf transpiration. Consequently, the RH was systematically higher at high light intensity, in sun plants, and amphistomatous leaves (Table S1) because the g_{sw} was typically greater in these conditions. In all treatment conditions, the final RH ($7.62\% - -20.4\%$) and VPD ($2.5\,\text{kPa} - -2.92\,\text{kPa}$) elicited rapid stomatal response and is comparable to treatments used in similar experiments (33, 53, 54, 78), but was

Table 2: Total number of humidity response curves analyzed in this study broken down by treatments.

	Measurement light intensity				
	Low		High		
Growth light intensity	amphi	pseudohypo	amphi	pseudohypo	Total
shade	275	275	270	266	1,086
sun	245	245	274	274	1,038
Total	520	520	544	540	2,124

not intended to mimic natural environmental variability. We started logging data after the transient WWR and continued logging data until g_{sw} reached its nadir. We then acclimated the leaf to low light (PPFD $= 150\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$) and RH $= 70\%$ before inducing stomatal closure with low RH and logging data again.

Prior to analysis, we removed unreliable and unusable data points, removed points not at instrument steady state, removed hysteretic portion of each curves at low g_{sw}, removed outliers within each curve, removed replicates with no overlap between amphi and pseudohypo curves from the same leaf, and thinned redundant data points within each curve. The rationale and procedure for each cleaning step is described in (30). Additionally, we excluded stomatal kinetic parameter estimates from four curves where τ was estimated to be greater than $1097\,\text{s}$, greatly exceeding the typical estimates (see Results). After these steps, the total number of humidity response curves analyzed is given in Table 2. The median number of curves per population per treatment was 9 and the median number of points per curve was 26. We extracted the the posterior distribution for τ and λ from each fitted curve to calculate the parameter estimate (median) and uncertainty (standard deviation) for subsequent analyses.

Stomatal anatomical traits l_{gc} and f_{gmax}

We estimated the stomatal density and l_{gc} on ab- and adaxial surfaces from all leaves used for humidity response curves. We used standard methods described fully in (30). For each surface we calculated the anatomical maximum stomatal conductance (g_{max}) at a reference leaf temperature of 25 °C following (57) as:

$$g_{\text{max}} = bmdl_{\text{gc}}/\sqrt{2}.$$

The biophysical and morphological constants b and m are:

$$b = \frac{D_{\text{wv}}}{v}, \text{ and}$$

$$m = \frac{\pi c^2}{j^{0.5}(4hj + \pi c)},$$

where D_{wv} is the diffusion coefficient of water vapor in air and v is by the kinematic viscosity of dry air. We assumed $D_{\text{wv}} = 2.49 \times 10^{-5}\ \text{m}^2\ \text{s}^{-1}$ and $v = 2.24 \times 10^{-2}\ \text{m}^3\ \text{mol}^{-1}$ (79). For kidney-shaped guard cells like those in wild tomatoes, $c = h = j = 0.5$. The $g_{\text{max,ratio}}$ is the g_{max} of the adaxial surface divided by the sum total g_{max} of both surfaces. To estimate f_{gmax}, we divided the initial g_{sw} estimated from the parameter g_{i} in Equation 1 from each humidity response curve by the anatomical g_{max} for that leaf. The estimated g_{i} was very close to the maximum g_{sw} observed near the beginning of each curve (Figure S2), suggesting that the initial g_{sw} was a good proxy for operational stomatal conductance at the time of the step change in humidity. We used the anatomical g_{max} for the surface(s) measured in each curve to calculate f_{gmax}. For amphi curves, we used the sum of adaxial and abaxial g_{max}; for pseudohypo curves, we used only the abaxial g_{max}.

Estimating kinetic parameters

We estimated τ and λ for each response curve by fitting Equation 1 to a time-series of g_{sw} values using Bayesian nonlinear regression. All models were fit using a Bayesian HMC sampling in the probabilistic programming language *Stan* (80) using the *R* package **brms** version 2.23.0 (81). We used *CmdStan* version 2.38.0 and **cmdstanr** version 0.9.0 (82) to interface with *R* version 4.5.2 (83). We sampled 1000 post-warmup draws from the posterior distribution on a single chain. We adjusted the number of sampling iterations and thinning interval for each curve so that the Gelman-Rubin convergence statistic ($\hat{R}$) (84) was less than 1.05 and the bulk effective sample size (ESS) was greater than 400 for all model parameters, and there were fewer than 10 divergent transitions. To answer Question 1, we assessed model fit using Bayesian R^2 (85) implemented in **brms**.

Estimating effects of stomatal anatomy on kinetic parameters

We first used model selection to determine whether anatomy (l_{gc} and f_{gmax}) affected kinetic parameters (λ and τ; Question 2) and at what level of biological organization (Question 3). From the selected model, we used parameter estimates and confidence intervals to test the predictions of our hypotheses. We interpreted nonzero fixed effects of l_{gc} and f_{gmax} on λ and τ as evidence that these anatomical traits influence variation in stomatal kinetics among individuals. We interpreted nonzero phylogenetic correlations among traits as evidence that these traits covary among populations. Finally, we used mediation analysis to test whether l_{gc} and/or f_{gmax} contribute to plasticity in stomatal kinetics. Table 3 summarizes the statistical criteria for for determining whether and at what level of biological organization stomatal anatomical traits affected kinetic parameters. The subsections below describe specific procedures for deciding whether those are met.

Table 3: Criteria used to determine whether and at what level of biological organization stomatal anatomical traits affected kinetic parameters.

Level of organization	Criteria supporting an association between stomatal anatomy and kinetics
Among individuals	• Anatomical parameters as fixed effects improve model fit (lower LOOIC) • Anatomical effects on kinetics are in the predicted direction • Confidence intervals of anatomical effects do not overlap zero
Plasticity among individuals	• Plasticity in anatomy mediates plasticity in stomatal kinetics
Among populations	• Confidence intervals of phylogenetic correlation among traits do not overlap zero • Phylogenetic covariance is in the predicted direction

Bayesian multiresponse phylogenetic models

Multiresponse models enable us to disentangle whether traits are associated among populations, individuals, and/or treatments that induce plasticity. The response variables in our models are τ, λ, l_{gc}, and f_{gmax}. All models included a subset of explanatory variables that were integral to the experimental design or necessary to account for statistical nonindependence. As explanatory variables, we included fixed effects of manipulative treatments: growth light intensity (sun and shade); measurement light intensity (high and low); and leaf type (amphi and pseudohypo). All models also included population as both a phylogenetically structured random effect and a non-phylogenetic random effect following (72). Prior to fitting, the phylogenetic (co)variance matrix was rescaled as a correlation matrix (86). For traits measured multiple times within the same leaf (f_{gmax}, λ, τ) we included a random effect of individual to account for repeated measures. We accounted for uncertainty in τ and λ using the standard deviation of the posterior distribution of each estimate (see 'Humidity response curves' for details). We modeled residual (co)variance, a combination among-individual (co)variance and measurement error, as a multivariate Student t-distribution, which is more robust to extreme values than the multivariate normal distribution (87). To account for heteroskedasticity, bounded measurements, and to linearize relationships, l_{gc}, λ, and τ were log-transformed prior to analysis whereas f_{gmax} was logit-transformed. All models were fit with *Stan* following the same methods described above for fitting Equation 1 to the humidity response curves.

Phylogenetic correlation among traits

We inferred that traits covary among populations if the 95% confidence intervals of the phylogenetic

correlation between traits did not overlap zero and the correlation was in the predicted direction. For example, a positive phylogenetic correlation between l_{gc} and τ would be consistent with the prediction that species with larger guard cells have slower stomatal kinetics. Phylogenetic covariance specifically indicates long-term evolutionary covariation among traits because our model simultaneously accounts for trait associations among individuals, shared plastic responses, and residual covariance. To test for possible causation, we used the inferred phylogenetic trait covariance matrix to estimate partial correlations between anatomical and kinetic variables (72). We interpret a significant partial correlation between variables as a plausible causal hypothesis.

Model selection and parameter estimation of individual-level effects

To infer whether individual-level variation in f_{gmax} and l_{gc} influence kinetic parameters, we used the leave-one-out cross-validation information criterion (LOOIC) to compare the fit of models using the *R* package **loo** version 2.9.0 (88) to calculate LOOIC values. We compared models with all permutations of f_{gmax} and l_{gc} influencing λ and τ. We considered all models within two ΔLOOIC standard errors of the minimum LOOIC (Table 4) to be in the set of plausible models to generate posterior predictions for hypothesis testing. Within each plausible model, we assessed whether an effect was significantly different from zero based on the 95% confidence intervals.

Mediation analysis to test for plastic effects

We used causal mediation analysis (89) to test whether plasticity in τ was mediated by plasticity in f_{gmax} or l_{gc} in response to measurement light intensity, growth light intensity, and curve-type treatments. We inferred the direct effect of treatments on τ from model coefficients. We estimated the mediated effect of treatments using the product of the effect of treatment on mediator (f_{gmax} or l_{gc}) and the effect of the mediator on τ (90). We determined significant direct and mediator effects based on whether the 95% confidence intervals of the effects were different from zero. We calculated effect sizes as the point estimate (median of the posterior) divided by the standard error (standard deviation of the posterior).

Variance decomposition and phylogenetic heritability

One explanation from why traits have affects at different levels of organization is that variance in those traits are concentrated at those different levels of organization. To evaluate this, we decomposed the variance in response variables (λ, τ, f_{gmax}, l_{gc}) into phylogenetic, population (nonphylogenetic), and among-individual levels. The phylogenetic variance component quantifies trait evolution between populations congruent with the phylogenetic relationships, whereas the population component quantifies variance among populations independent of phylogenetic relationships (86, 91). We estimated these components from the associated random-effect standard deviations for each response variable. The among-individual variance components quantify variance among individuals within the same population and treatment. We estimated the among-individual component as the sum of the individual random-effect and residual standard deviations. It was not possible to estimate individual random-effect in l_{gc}

with our study design because we only measured average guard cell length once per leaf. The among-individual component is biased upwards because it also includes measurement error, which cannot be separately estimated with our experimental design. The phylogenetic heritability is equivalent to the phylogenetic variance component and is estimated following (91); (86) as:

$$h^2_{\text{phy}} = \frac{\sigma^2_{\text{phy}}}{\sigma^2_{\text{phy}} + \sigma^2_{\text{pop}} + \sigma^2_{\text{ind}}},$$

where σ^2_{phy}, σ^2_{pop}, σ^2_{ind} are the phylogenetic, population (nonphylogenetic), and among-individual variance components, respectively.

Effect of adaxial stomata on kinetics

If adaxial stomata respond faster than abaxial stomata (Question 4), we predicted that τ should be greater in pseudohypo leaves compared to amphi. We evaluated this prediction by estimating the fixed effect of leaf type (amphi vs pseudohypo) on τ in the multiresponse model.

Results

The time-constant τ explains most variation stomatal closure kinetics

Stomatal conductance (g_{sw}) decreased rapidly in response to a step change in humidity. The model in Equation 1 fit the data extremely well (Figure S1). The mean and minimum Bayesian R^2 across all curves was 0.9994 and 0.9802, respectively. Consider a typical leaf in the experiment characterized by the median time constant (τ) and lag time (λ) among wild tomato accessions in all treatments. After humidity decreased and the transient wrong way response elapsed, it took on average 127 s for g_{sw} to decrease half-way (t_{50}) from its initial (g_{i}) to final (g_{f}) steady state value. In terms of the kinetic model parameters, most variation among leaves was due to difference in the time constant rather than the lag time. In the previous example, increasing λ from its median to its maximum estimated value among accessions increased the t_{50} by only 7.1 s. In contrast, increasing τ from its median to maximum value increased t_{50} by 88 s. Hence, we primarily focus on treatments and anatomical parameters affect τ.

Variation in guard cell length and aperture

Guard cell length (l_{gc}) differed among species and varied to a lesser extent between growth light intensity treatments and leaf surfaces. In contrast, stomatal conductance as a fraction of maximum conductance (f_{gmax}) was most strongly determined by measurement light intensity. Excluding treatment effects, phylogenetically structured differences among populations (i.e. phylogenetic heriability) accounted for 64.4% [95% CI: 32.3 to 81.0%] of the variance in l_{gc} (Figure 1, Table S2). There was also developmental plasticity in l_{gc}. On average, l_{gc} increased by 12.1% [95% CI: 11.1 to 13.0%] in the sun

treatment (Figure 2a; Table S5). The average l_{gc} of pseudohypo leaves was -5.8% [95% CI: -6.6 to -5.1%] shorter than amphi leaves (Figure 2a; Table S5). This was not because of plasticity, as we measured amphi and pseudohypo curves on the same leaf. Rather, this was a composition effect. Adaxial stomata are larger than abaxial stomata in these species (Figure S3), so the average l_{gc} in pseudohypo leaves was higher than in amphi leaves because the former only included abaxial stomata.

The phylogenetic heritability of f_{gmax} was low, accounted for 16.6% [95% CI: 3.1 to 34.9%] of the variance (Figure 1, Table S2). High measurement light intensity increased f_{gmax} by an average of 0.85 logit units [95% CI: 0.82 to 0.87] (Figure 2b; Table S5). For example, if $f_{gmax} = 0.1$ in low light, the typical f_{gmax} for the same leaf acclimated to high light was 0.21. Sun leaves had on average -0.28 logit units [95% CI: -0.35 to -0.21] lower f_{gmax} than shade-grown plants (Figure 2b; Table S5). Treated pseudohypo leaves had on average 0.21 logit units [95% CI: 0.18 to 0.24] higher f_{gmax} than untreated, amphistomatous leaves (Figure 2b; Table S5). This indicates that stomatal aperture was more open in leaves where gas exchange through adaxial stomata was blocked.

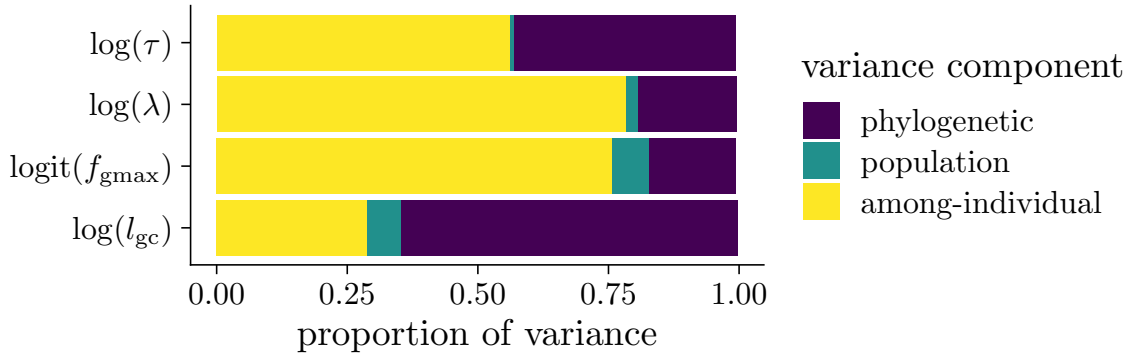

Figure 1: **Traits vary at different levels of biological organization**. A preponderance of the variation in guard cell length (l_{gc}) and the time-constant τ is phylogenetically structured among populations. In contrast, most of the variance in stomatal conductance as a fraction of anatomical maximum stomatal conductance (f_{gmax}) and the lag-time (λ) is among individuals. Nonphylogenetic variation among populations was low for all traits.

Variation in stomatal closure kinetic parameters

Estimates of stomatal closure kinetic parameters for each curve are in Table S3 and estimates for each population are in Table S4. These will be deposited on Dryad upon publication. The time-constant τ exhibited moderate phylogenetic heritability, accounting for `rtau_phylo_var‘` of the variance, whereas λ varied more within populations (Figure 1, Table S2). Some light treatments also affected kinetic parameters. Growth light intensity did not significantly affect τ (Figure 2, Table S5). In sun plants, τ was on average 5.59% [95% CI: 0.993 to 10.3%] lower than in shade plants after accounting for other explanatory variables. Within the same leaf, increasing the measurement light intensity prior to the humidity step change from 150 μmol m^{-2} s^{-1} to 2000 μmol m^{-2} s^{-1} significantly increased τ by

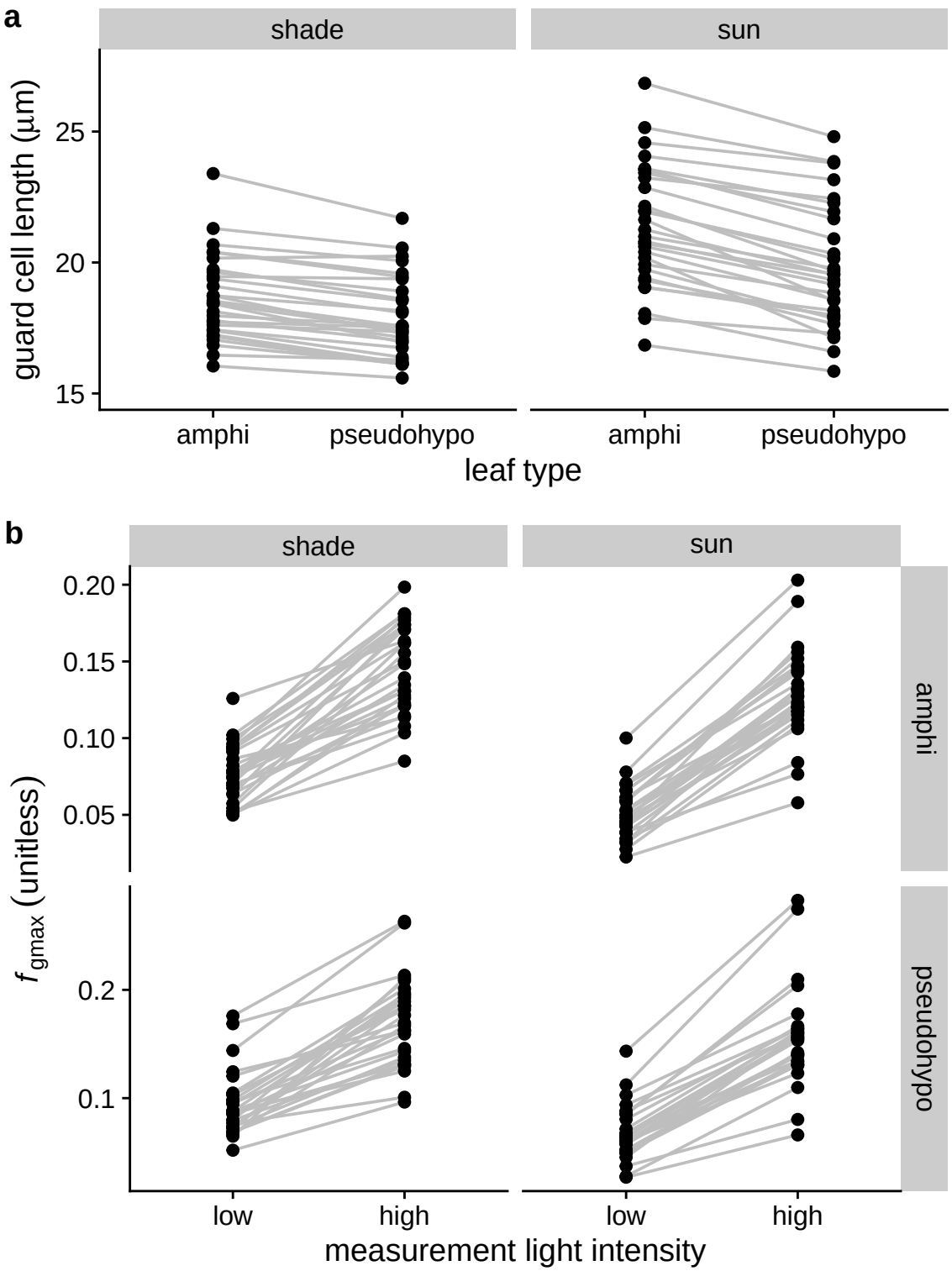

Figure 2: Caption on following page.

17.8% [95% CI: 11.6 to 24.7%] (Figure 2, Table S5). The time lag λ responded to growth, but not measurement light intensity after accounting for other explanatory variables (Figure 2, Table S5). In sun plants, τ was on average 12.1% [95% CI: 10.2 to 14.1%] greater than in shade plants. Higher measurement light decreased increased λ slightly by 1.2% [95% CI: -1.13 to 3.62%].

somewhere, random effect Table S6

Guard cell size and f_{gmax} affected stomatal kinetics, but at different levels of biological organization. In the model with the lowest LOOIC (see Table 4 for model comparison results), the only significant partial correlation was between l_{gc} and τ ($\rho = 0.72$ [95% CI: $-0.073 to 0.97$)] was a significant phylogenetic correlation between l_{gc} and τ (**fig?-**?). PARTIAL CORRELATION RESULT. Variation in l_{gc} within populations did not explain within-population variation in τ Table S5. NOTE THAT BEST MODEL RETAINED INDIVIDUAL LEVEL EFFECT OF GCL, BUT COEFFICIENT WAS NOT SIGNIFICANT. OTHER BEST MODELS DID NOT RETAIN THIS. DID GCL MEDIATE AFFECT OF TREATMENTS?

Need to show (this is a growing, incomplete list): - phylogenetic partial correlation - partial correlation showing that gcl and tau are (only?) significant phylogenetic covariates

Stomatal aperture mediates plasticity in τ

We imposed three treatments that influenced stomatal kinetics: growth light intensity, measurement light intensity, and leaf type. All three treatments directly effect τ and had indirect effects mediated by f_{gmax} (Figure 4). We did not estimate effects mediated by l_{gc} because the selected model did not retain direct effects of l_{gc} on τ (Table 4). The direct effect of the high light intensity (sun) growth treatment increased τ by 5.57% [95% CI: 0.99 to 10.33%], but the indirect of lower f_{gmax} counteracted by -5.09% [95% CI: -7.36 to -3.22%]. In contrast, direct and indirect effects of high measurement light intensity increased τ. The direct effect increased τ by 17.81% [95% CI: 11.56 to 24.74%] and the indirect effect mediated by f_{gmax} increased τ by 17.33% [95% CI: 11.23 to 23.73%]. The direct effect of the pseudohypo treatment decreased τ slightly by -2.63% [95% CI: -5.03 to -0.09%], but the indirect effect mediated by greater f_{gmax} increased τ by 4.03% [95% CI: 2.58 to 5.53%].

Discussion

The ability for stomata to adjust to fluctuating environmental conditions likely evolved in early land plants because of selection to optimize water use relative to carbon gain (2). Hydroactive control of stomatal aperture in angiosperms increases flexibility to modulate stomatal aperture depending on leaf water status (5), perhaps enabling greater maximum hydraulic conductance and photosynthesis [mcadam?]. We now understand the steady-state response of stomatal conductance (g_{sw}) to global increases in vapor pressure deficit (VPD) better but there is significantly greater uncertainty surrounding kinetic responses to VPD (28). We considered two nonmutually exclusive mechanistic hypotheses for how stomatal anatomy influences stomatal closure kinetics. First, smaller stomata may close faster because of greater surface area to volume ratio (42). Second, stomatal responses should be slower in

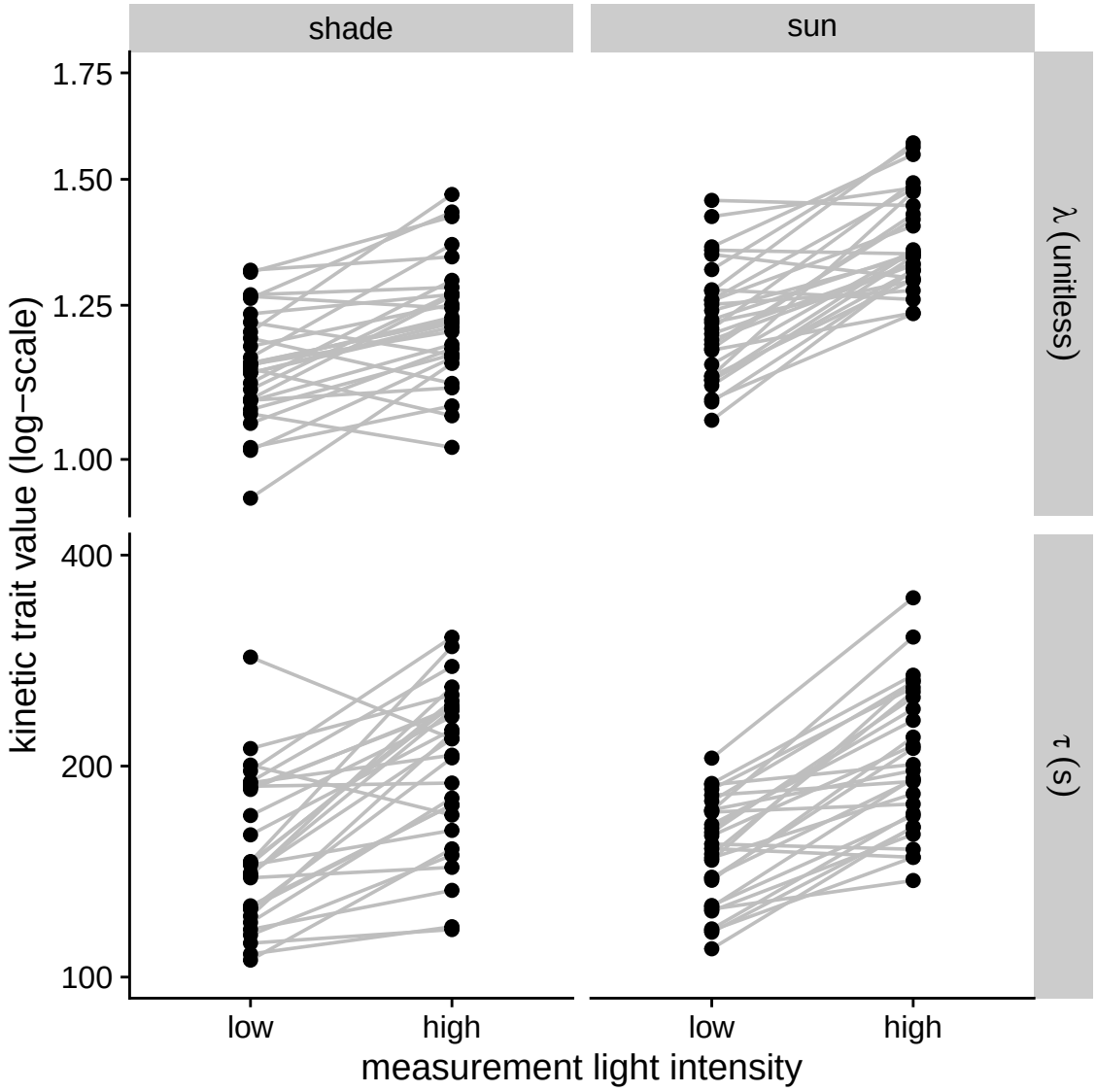

Figure 2: **Stomatal kinetics parameters λ (lag time; top facets) and τ (time constant; lower facets) vary among wild tomato populations.** Stomatal conductance decreases faster (lower τ) in response to a step change in vapor pressure deficit (VPD) in leaves when measured under low light intensity. The pattern was consistent for both shade (left facets) and sun (right facets) grown plants. The pattern for λ was qualitatively similar to that for τ. Each point is the average parameter value for one accession in that treatment combination. The growth and measurement light intensity treatments are described in the Materials and Methods section.

Table 4: Model comparison results. Models with a ΔLOOIC less than two standard errors (SE) from the minimum LOOIC are considered to have substantial support and are used for posterior predictions to test hypotheses about the effects of guard cell length (l_{gc}) and the stomatal conductance as a fraction of anatomical maximum stomatal conductance (f_{gmax}) on stomatal kinetic parameter λ (time lag) and τ (time constant). Unsupoorted models are in gray italic font. The model selected for subsequence inference is in bold. LOOIC stands for leave-one-out cross-validation information criterion.

		λ		τ	
ΔLOOIC	SE	f_{gmax}	l_{gc}	f_{gmax}	l_{gc}
0.00	0.00	✓		✓	✓
3.51	2.56	✓	✓	✓	✓
3.57	**2.60**	✓		✓	
4.17	2.74	✓	✓	✓	
19.58	10.66	✓			✓
21.19	10.71	✓			
23.84	*10.76*	✓	✓		✓
24.18	*10.70*	✓	✓		
47.84	*14.41*			✓	✓
50.07	*14.50*		✓	✓	
50.49	*14.48*		✓	✓	✓
56.07	*14.52*			✓	
68.18	*18.48*				
69.96	*18.31*		✓		
74.42	*18.31*		✓		✓
75.56	*18.29*				✓

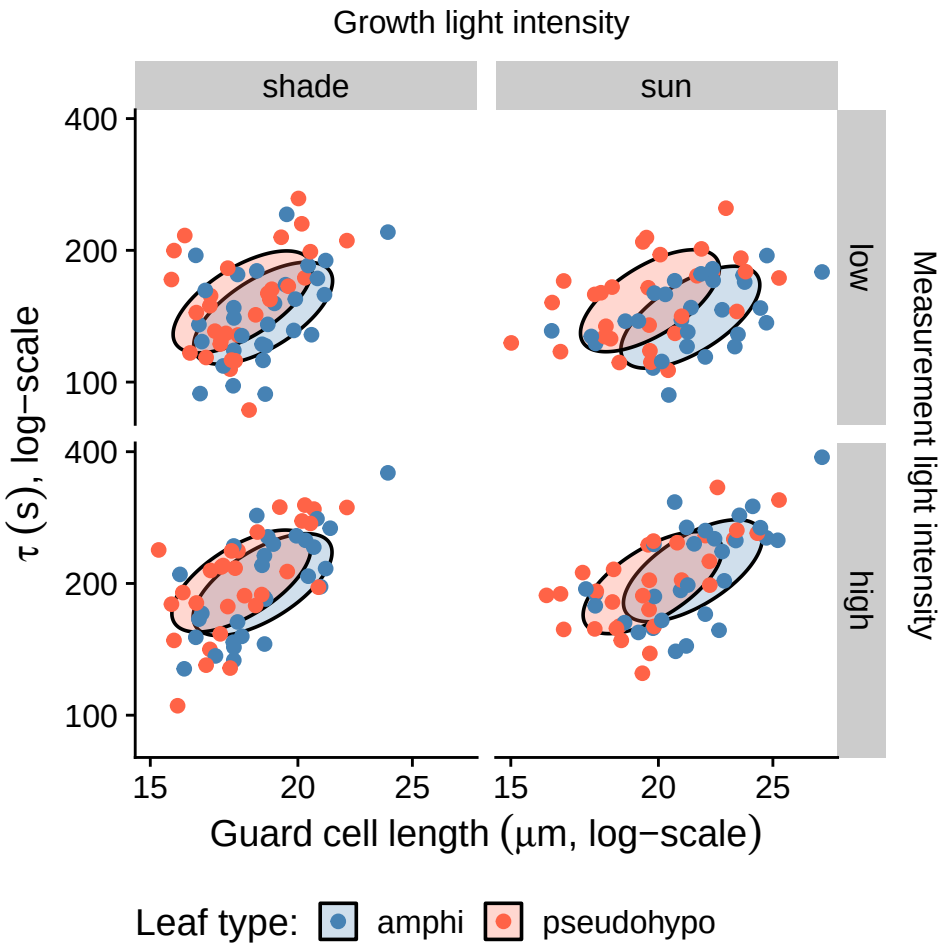

Figure 3: **Phylogenetic covariance between guard cell length (l_{gc}, x-axis) and the stomatal closure time-constant (τ, y-axis) indicates that populations with larger stomata tend to closer more slowly in response to humidity.** The overall pattern was consistent across grow light intensity treatments (left and right facets), measurement light intensity treatments (top and bottom facets), and leaf types (point colors). The direction and magnitude of the ellipses are such that they should encompass 95% of the phylogenetically structured (co)variance in l_{gc} and τ on log-transformed scales. The location of ellipses are set to the median trait values in each treatment combination for visual comparison. Each point is the average parameter value for one accession in that treatment combination. The growth and measurement light intensity treatments are described in the Materials and Methods section.

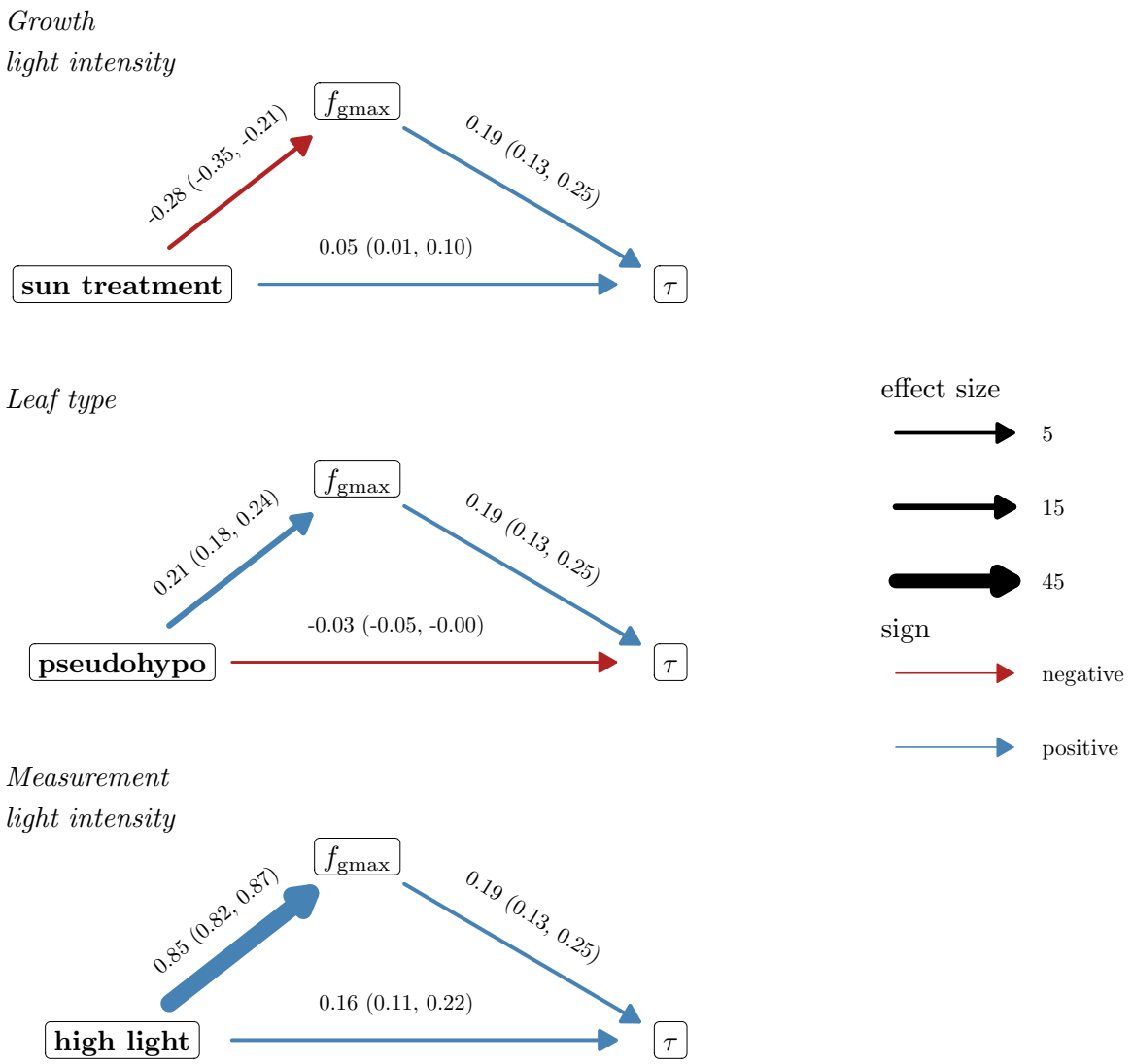

Figure 4: **Stomatal aperture as a fraction of anatomical maximum stomatal conductance (f_{gmax}) mediates plasticity in the time constant τ.** The three treatments are leaf type (top facet; amphi vs. pseudohypo), growth light intensity (middle facet; shade vs. sun), and measurement light intensity (bottom facet; low vs. high). Horizontal arrows at the bottom of each graph indicate the direct effect of the treatment level on τ; slanted arrows leading to or from f_{gmax} indicate mediated effects of the treatment on τ. Line thickness is proportional to the effect size and line color indicates whether the effects are negative (red) positive (blue). Parameter estimates of the effect and 95% confidence intervals are above each line. Note that are not consistent because f_{gmax} is on a logit scale whereas τ is on a log-seconds scale. The selected model did not retain direct effeects of l_{gc} on τ and hence this variable could not be considered as a mediator.

guard cells under higher turgor pressure (59). Based on these hypotheses, we predicted that leaves with smaller guard cells and operating further from their anatomical maximum stomatal conductance (i.e., lower f_{gmax}) would have faster stomatal closure kinetics (lower τ). Analysis of 2,124 time courses from 29 populations in multiple environments, …

note: this paper (48) also found relationship between gop and kinetics

effect of fgmax is not simply because high light intensity increased fgmax and tau. Causal inference dissects different causal paths and shows direct effect of fgmax on tau within populations and treatments

caveat: this treatment is a hammer than exposes maximum rate of stomatal closure. The factors which influence this may not explain variance in kinetics in more natural conditions.

Need to mention subsaturation. If airspace is unsaturated, then gs is underestimated. This doesn't necessarily affect kinetic parameter estimates since all measurements were done at high VPD.

(not sure where this goes) It might seem that natural selection should favor faster stomatal responses to get closer to the ideal of instantaneous optimization. Under this assumption, variation in stomatal kinetics would depend on how strong directional selection for faster responses is counterbalanced by costs of faster stomata. We discuss multiple, nonmutually exclusive costs below [WHERE?]. However, once models account for inevitable time lags, natural selection can favor slower responses when this prevents overreacting to unpredictable and shortlived changes in the environment. [Review some lit] It is therefore likely that the direction of selection on stomatal response rate varies depending on the predictability and duration of environmental fluctuations, as well as the energetic costs of stomatal movements.

Traditional view is that smaller stomata are selected for speed and because there is higher gmax for same fs (Franks). Our results suggest an alternative hypothesis. Smaller stomata may increase rate of stomatal closure (we haven't tested opening), but only if fgmax is low or moderate. Therefore if leaves with smaller stomata maintain a similar gop, then stomatal density must increase to prevent fgmax from increasing. This may explain the association between the association between small stomata and high density, and relatedly the dearth of leaves with sparse, small stomata.

Might be useful to relate to literature on clines in stomatal size, especially poplar (92) and (93) and (94)

note about adaxial stomata: Mott (2007) reported that stomatal responses to humidity were independent on the two surfaces of amphistomatous leaves of Vicia faba and Xanthium strumarium: when humidity was changed at one surface only, stomata on that surface responded but stomata on the other surface did not." (quote from Buckley et al 2011). FYI, tomatoes have BSEs (heterobaric)

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Table S1: Relative humidity (RH) and leaf vapor pressure deficit (VPD) during stomatal kinetics measurements systematically differed because of differences in stomatal conductance. Stomatal conductance tended to be greatest in sun-grown plants, measured a high light intensity, and untreated (amphistomatous). The mean values across replicate curves in each treatment conditions are given; within each curve we calculated the median RH and VPD across time points before calculating the mean among curves.

[!h]

Growth light intensity	Measurement light intensity	Leaf type	RH (%)	VPD (kPa)
shade	low	pseudohypo	7.62	2.92
shade	low	amphi	8.25	2.91
shade	high	pseudohypo	13.3	2.74
shade	high	amphi	14.7	2.69
sun	low	pseudohypo	9.63	2.85
sun	low	amphi	11.1	2.81
sun	high	pseudohypo	16.8	2.62
sun	high	amphi	20.4	2.50

Supplementary Materials

Supporting tables

Table S2: The relative contribution of phylogenetic, among population (nonphylogenetic), and among-individual varation differs among stomatal traits. The phylogenetic component is equivalent to the phylogenetic heritability. The population component is the non-phylogenetic variance among populations. The among-individual variance component is the variation among individuals within a population after accounting for treatment effect. Estimates and 95% confidence intervals (CI) are estimated as the median and quantile intervals of the posterior distribution. f_{gmax}: stomatal conductance as a fraction of anatomical maximum stomatal conductance; l_{gc}: guard cell length; λ: lag time; τ: time constant.

component	% variance	95% CI
$\log(l_{\mathrm{gc}})$		
phylogenetic	64.4	[32.3, 81.0]
population	6.6	[0.9, 25.7]
among-individual	28.7	[17.0, 45.0]
$\mathrm{logit}(f_{\mathrm{gmax}})$		
phylogenetic	16.6	[3.1, 34.9]
population	7.0	[2.1, 17.2]
among-individual	75.7	[59.6, 85.5]

Table S3: : Raw data file in CSV format containing estimates of stomatal anatomy and kinetic variables associated with each curve. These are raw data for fitting multiresponse models. A final version will be deposited on Dryad after acceptance. This version is available for reviewers.

Table file:
Download from:
https://github.com/cdmuir/solanum-kinetics/raw/refs/heads/main/tables/tbl-estimates-curve.csv

(continued)

component	% variance	95% CI
$\log(\lambda)$		
phylogenetic	18.9	[4.8, 35.9]
population	2.3	[0.1, 9.9]
among-individual	78.3	[62.2, 88.3]
$\log(\tau)$		
phylogenetic	42.4	[24.2, 58.9]
population	1.0	[0.0, 7.7]
among-individual	56.1	[40.2, 70.6]

Table S5: Fixed effect parameter estimates and 95% confidence intervals (CIs) from the posterior distribution of the selected model. For each response variable, we estimated effects of growth light intensity (sun vs. shade), measurement light intensity (high vs. low), and leaf type (amphi vs. pseudohypo). The selected model potentially includes of effects of l_{gc} and f_{gmax} on τ and λ.

Parameter	**Estimate [95% CI]**
$\log(l_{\mathbf{gc}})$	
intercept (shade, low light, amphi)	2.96 [2.79, 3.13]
effect of sun growth treatment on $\log(l_{\mathrm{gc}})$	0.11 [0.11, 0.12]
effect of pseudohypo leaf type on $\log(l_{\mathrm{gc}})$	-0.06 [-0.07, -0.05]
$\mathrm{logit}(f_{\mathbf{gmax}})$	

Table S4: : Raw data file in CSV format containing estimates of stomatal anatomy and kinetic variables associated with each population based on multiresponse model predictions. A final version will be deposited on Dryad after acceptance. This version is available for reviewers.

Table file:
Download from:
https://github.com/cdmuir/solanum-kinetics/raw/refs/heads/main/tables/tbl-estimates-accession.csv

(continued)

Parameter	**Estimate [95% CI]**
intercept (shade, low light, amphi)	-2.50 [-2.79, -2.19]
effect of sun growth treatment on logit(f_{gmax})	-0.28 [-0.35, -0.21]
effect of high measurement light intensity on logit(f_{gmax})	0.85 [0.82, 0.87]
effect of pseudohypo leaf type on logit(f_{gmax})	0.21 [0.18, 0.24]
log(λ)	
intercept (shade, low light, amphi)	0.39 [0.28, 0.48]
effect of sun growth treatment on log(λ)	0.11 [0.10, 0.13]
effect of high measurement light intensity on log(λ)	0.01 [-0.01, 0.04]
effect of pseudohypo leaf type on log(λ)	-0.04 [-0.05, -0.03]
effect of logit(f_{gmax}) on log(λ)	0.10 [0.08, 0.13]
log(τ)	
intercept (shade, low light, amphi)	5.50 [5.14, 5.84]
effect of sun growth treatment on log(τ)	0.05 [0.01, 0.10]
effect of high measurement light intensity on log(τ)	0.16 [0.11, 0.22]
effect of pseudohypo leaf type on log(τ)	-0.03 [-0.05, -0.00]
effect of logit(f_{gmax}) on log(τ)	0.19 [0.13, 0.25]

Table S6: Random effect parameter estimates and 95% confidence intervals (CIs) from the posterior distribution of the selected model. For each response variable, we estimated among-individual random effects from repeated measures of the same individual, where possible, random effects of population (nonphylogenetic), and phylogenetically structured random effects of population. We report the random effect standard deviations (SD). For population-level random effects, we also estimated correlations between traits.

Parameter	**Estimate [95% CI]**
among-individual random effects	
SD in log (λ)	0.08 [0.07, 0.08]
SD in log (τ)	0.21 [0.19, 0.23]
SD in logit $\left(f_{\text{gmax}}\right)$	0.38 [0.35, 0.41]
phylogenetic random effects	
SD in log (λ)	0.07 [0.02, 0.11]
SD in log (τ)	0.27 [0.16, 0.40]
SD in log $\left(l_{\text{gc}}\right)$	0.14 [0.07, 0.21]
SD in logit $\left(f_{\text{gmax}}\right)$	0.23 [0.07, 0.40]
correlation between in log (λ) and log (τ)	0.45 [-0.18, 0.84]
correlation between in log (λ) and log $\left(l_{\text{gc}}\right)$	0.11 [-0.48, 0.70]

(continued)

Parameter	Estimate [95% CI]
correlation between in log (λ) and logit (f_{gmax})	-0.01 [-0.63, 0.63]
correlation between in log (τ) and log (l_{gc})	0.67 [0.16, 0.91]
correlation between in log (τ) and logit (f_{gmax})	0.21 [-0.46, 0.72]
correlation between in log (l_{gc}) and logit (f_{gmax})	0.47 [-0.26, 0.88]
population (nonphylogenetic) random effects	
SD in log (λ)	0.02 [0.00, 0.05]
SD in log (τ)	0.04 [0.00, 0.12]
SD in log (l_{gc})	0.04 [0.01, 0.08]
SD in logit (f_{gmax})	0.15 [0.07, 0.26]
correlation between in log (λ) and log (τ)	0.07 [-0.75, 0.80]
correlation between in log (λ) and log (l_{gc})	0.04 [-0.68, 0.74]
correlation between in log (λ) and logit (f_{gmax})	-0.23 [-0.82, 0.55]
correlation between in log (τ) and log (l_{gc})	0.27 [-0.67, 0.89]
correlation between in log (τ) and logit (f_{gmax})	0.11 [-0.69, 0.78]
correlation between in log (l_{gc}) and logit (f_{gmax})	0.40 [-0.27, 0.84]

Supporting figures

Figure file:
Download from:
https://github.com/cdmuir/solanum-kinetics/raw/refs/heads/main/figures/rh-curves.pdf

Figure S1: : Humidity-response curves and fitted lines. The title provides the Tomato Genetics Resource Center accession number, replicate letter, and species names. The subtitle indicates the growth light intensity (sun or shade), measurement light intensity (low or high), and leaf type (amphi or pseudohypo). Points are raw data and lines are fitted curves. The Bayesian correlation coefficient, Bayes R^2, is shown to the right of each curve.

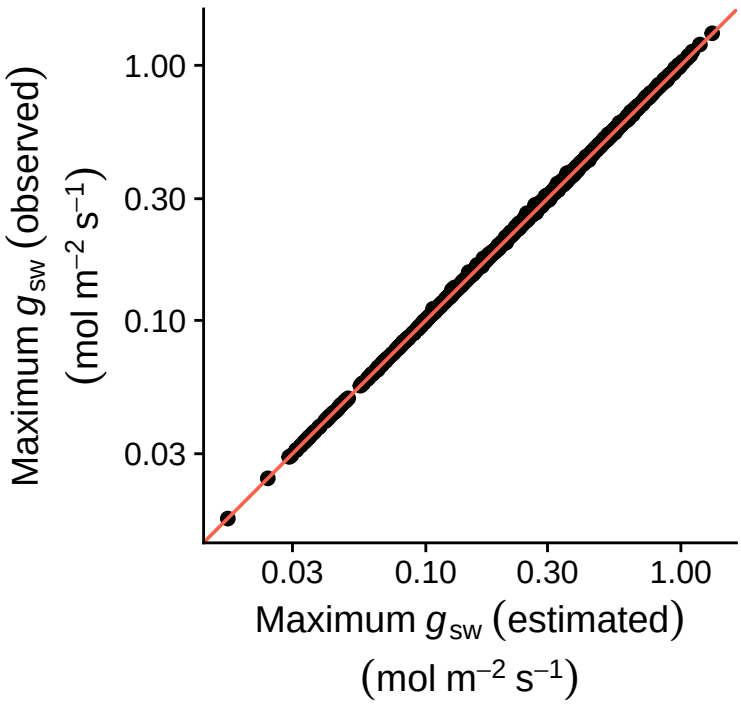

Figure S2: **The estimated (x-axis) and observed (y-axis) maximum stomatal conductances (g_{sw}) are nearly identical.** Each point is the empirical maximum g_{sw} plotted against the estimated g_{sw} for every humidity response curve. The orange line is the 1:1 line for reference. Both axes are on a log-scale.

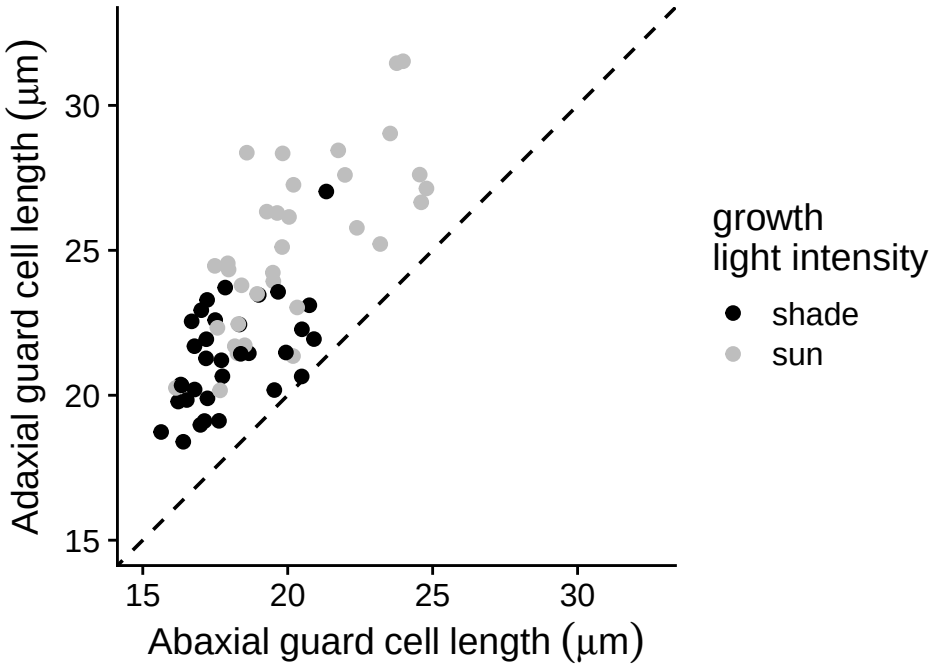

Figure S3: **Adaxial (upper) stomata are consistently larger than abaxial (lower) stomata in both sun-grown (grey points) and shade-grown (black points) plants.** Each point was calculated by taking the median guard cell length within each biological replicate and then calculating the mean among replicates within each population and growth light intensity treatment. The dashed line is the 1:1 line for reference. All points above this line indicate larger values on the adaxial surface.